L5.5: What is a linear transformation

Transcript Language

English

Transcriber's Name: Crescendo Transcriptions Pvt Ltd

Mathematics of Data Science 2 Professor Sarang Sane

Department of Mathematics, Indian Institute of Technology, Madras

What is a linear transformation

Hello, and welcome to the Maths 2 component of the online degree on Data Science. In this

video, we are going to study what is a linear transformation.

So, let us just recall before we go ahead, what is a linear mapping, this is what we

studied in the previous video. A linear mapping f from Rn to Rm, so it is a function, and

the function has a very particular form. Namely, the form is that every coordinate is given

by a linear combination. This is the form of the function, and linear combination with

what coefficients, the coefficients are real numbers.

So, we studied this to some extent in the previous video, and we saw the use of such

a thing. For example, we looked at the examples of the grocery shops. And we noticed that

linear mappings have some very interesting properties called linearity. So further, we

can write this in the form of matrix multiplication, A times x, where the ijth entry of the matrix

A is little aij and X is the column with X1, X2, Xn, and linear mapping satisfy linearity,

which means if you have f of X1 plus Cy1, f of X2 plus Cy2 and so on and Xn plus Cyn

that is f of X1, X2, Xn plus C times f of Y1, Y2, Yn.

So, let us define what is a linear transformation. So, a linear transformation is basically whatever

we want the same properties that we had in the previous slide, but for arbitrary vector

spaces, instead of for Rn and RM we look at vector spaces V and W. So, we say, a function

f from V to W is a linear transformation, if for any two vectors, v1 and v2 in the vector

space V, and for any c in R so that is a scalar the following conditions hold, f of V1 plus

V2 is f of V1 plus f of V2, and f of C times V1 one is equal to C times f of V1.

So before going ahead, maybe, I will point out that this is the same as linearity. So,

this is equivalent to linearity as in the previous slide, which we said that f of V1

plus C times V2 is f of V1 plus C times f of V2. Why is that? Suppose we assume it has

linearity, then, of course, both of these are clear, because we can put C equals 1 and

that will give us the first condition.

So, by putting c equals one, we get this. By taking V1 to be zero and calling V2 as

V1, we get the second 1. So, if we know linearity is true, then we get both these that both

these conditions hold. Conversely, if we know both of the conditions hold, then if we have

V1 plus CV2, so I do not know linearity a priory, but I know that these two conditions

hold.

So, I can treat CV2 as a vector, and use the first condition to write this as a f of V1

plus f of C times V2. We use the first equation with CV2 in place of this V2 and V1 in place

of the V1. And then we use the second equation, the second condition for writing the second

term here, f of CV2 as C times f of V2, and that is what we want for linearity. So, this

is equivalent to linearity. So, you could either express it this way or you could express

it in terms of the way we have written it, which is we have written it earlier that is,

which is f of V1 plus CV2 is f of V1 plus C times f of V2.

Let us look at some examples. So, here are some examples. So, these are all going to be examples of linear mapping. So, in particular, they are linear transformations. So, I should

have pointed out before that linear mappings are linear transformations because linear

mapping satisfy linearity.

So maybe I should write here. So, linear mappings are linear transformations as we have defined

above. So, linear mappings mean when we have V and W, as some Rn and Rm respectively and

then the function f is given by a bunch of a collection of linear combinations. We have

seen already that in that case that satisfy linearity, and we have checked linearity is

the same as these two conditions.

So, all these examples are going to be examples of linear mapping. So, in particular, they

are linear transformations. So, f of x, y is 2x comma y, f of x, y is 2x comma 0, f

of x, y, z which is from R3 to R3 is x by 2, 3 y comma 5z. R3 to R4 this time, so f

of x, y, z is 4y minus z, 3y plus 11 by 19z, 5x minus 2z, 23y, and then maybe R to R3, three which is f of t is t comma 3, t comma 23 by 89 t.

And maybe a last function which is also linear transformation, which is a projection function

f of x, y is x, so it projects onto the x, x. We will see this later in some other context

a few weeks from now.

So, all these are linear mapping so in particular they are linear transformations. Of course,

we will also see some non-trivial examples of linear transformations. By non-trivial

I mean, where the V or W are not R2 or R3. I mean not Rm or Rn.

So, let us recall before going ahead what is a one-one function and what is an onto

function. Possibly you have seen this already Maths 1, if not, just look at this definition.

So, function f from V to W. Now V and W, are need not be vector spaces here these could

be any sets. So, it is called one-one or injective. If f of V1 is f of V2 implies V1 as V2.

Similarly, such a function is called onto or surjective if for any little w in capital

W. We have a little v and capital V, such that f of little v is little w. So, for a

linear transformation being one-one is equivalent to f of V0 implies V0. Why is that? So, let

us prove that.

So, let us assume that f is one-one. Notice f is a linear transformation. I am assuming

it is a linear transformation. Otherwise, of course, this is none of this even makes

sense because we do not even know what is the 0. So, f is one-one. So, f is a linear

transformation and f is one-one, assume f is one-one. So then, we know f of V1 is f

of V2. This is given to us.

But on the other hand, we know that f is a linear transformation. So, one of the properties

of a linear transformation is that f of 0 is 0. So, f of 0 is 0. So here what do we

mean by 0 and 0 here, this, the 0 inside the bracket is the 0 in V it is a vector space,

remember, so it has a 0 and 0 outside f of 0, the value is 0 in W. So, this is true because it is a linear transformation.

Maybe, I should point out why for a linear transformation f of 0 is 0 that is because

if you take f of 0 plus 0, then we get f of 0 plus f of 0, so this implies f of 0 is equal

to f of 0 plus f of 0. So, we can, so this is happening inside a vector space W, so we

can subtract f of 0 from both sides.

So, what that will give us is that f of 0 is 0. So, we know that, so this is true for

any linear transformation, f of 0 is 0. So now we want to check that f of V0 implies

a V0. So, if f of V is 0 that means f of V is the same as f of 0. And now I can use the

fact that f is one-one to conclude that V is 0. So, this shows that if f is one-one,

then this condition is satisfied that f of V0 implies V0.

So, one way we have shown one way that if it is one-one it satisfies this condition.

Conversely, so that means we assume that assume f of V0 implies V0. So now I want to prove it is one-one. So, let us assume that f of V1 is f of V2, but then that will mean that

f of V1 minus V2 is equal to 0.

So, again, here I have to explain why. So that is because I can subtract f of V2 on

both sides so I will get f of V1 minus f of V2 is equal to 0. But now, I can take this

minus inside. So, what I am claiming is that this is f of V1 plus f of minus V2. This is

0.

And then I am using a linearity because it is a linear transformation to say that V1,

f of V1 minus V2. So, the key step is to is prove that I can take minus inside. Why can

I take minus inside? Let me do that here. So, the reason is, the same argument that

we had before. So, if we have V plus minus V, then this is f of 0 is 0, because we just

proved f of 0 is 0.

So now you expand the first term, the first expression here, this expression, so you write this as f of V, plus f of minus V, and then you take the f of V on the other side, so

that will tell you that f of minus V is minus f of V, that is what we wanted to prove. So,

I have proved indeed, that f of V1 is f of V2, you subtract f of V2 on both sides and

then that will give you f of V1 minus V2. So, you get f of V1 minus V2 is 0. But now

I know this condition that f of V0 implies V0, that means V1 minus V2 is 0. That means

V1 is equal V2, adding V2 both sides. So, we have proved that they are equivalent.

So, in general, if you have a arbitrary function on sets, checking one-one means you have to

check this condition that f of V1 is f of V2 implies V1 is V2. But if you have a linear

transformation, you can instead just check that f of V0 implies V0. So, it reduces to

checking those for those values of V, which so that when you apply f to those values,

they become 0. So, this is a special set and we are going to study that set later. In fact,

it is a subspace.

So now let us get to what is an isomorphism, which is why we studied this notion of one-one

and onto. So, recall first that a function f, again, for any arbitrary sets, V and W,

if you have a function f from V to W it is called a bijection or we say that it is bijective.

If it is one-one and onto. So, or sometimes it is called a one-one correspondence. So,

it is one-one and onto. What that means is, what that means is, if you have V here, then

there is a unique W that corresponds to it, which is what we are calling f of V.

So, there is exactly one such V, such that f of V is equal to W. For each W such a V

exists, and it is unique, that is what is called a bijection or that is the essence

of a bijection. So in other words, you can pair up elements, you have V here, W here,

V here, W here, that those pairs are exactly, the W paired with V is what we are calling

f of V.

So that is, if you can do such a thing that is called a bijection. Fine, so what is an isomorphism? So, this is what I just said, bijection is equivalent to saying that for

any W in W there exists a unique V such that f of V is W. So, a linear transformation between

two vector spaces V and W is an isomorphism if it is a bijection. So, it is an isomorphism

if it is a bijection. So, it is a linear transformation along with a bijection. So it satisfies that

f of, it satisfies linearity, which means f of V1 plus Cv2 is f of V1 plus C times f

of V2 and it satisfies that it is one-one and onto.

Namely, that for every W and W, there is a unique little v in capital V, such that f

of V is little w. So, example one seen earlier is an isomorphism, let us work that out. So,

what was the example, it was f from R2 to R2 where f of x, y is 2x comma y, we already

know this is a linear transformation because it is a linear mapping.

So, to see that it is a bijection so f of x, y, so because it is a linear transformation

to check injectivity it is enough to check that whenever this becomes 0 and remember that 0 in R2 is the point on the vector 0, 0. That means, 2x comma y is 0, 0. How do

I get this? By evaluating f on x comma y so we know that is 2x comma y.

But when will this be 0 comma 0? This will be 0 comma 0 precisely when our 2x is 0, and

y is 0. So already, we have got that y is 0, but 2x is 0 means that x is 0 where all

the real numbers. So, I can see that, which means x comma y is 0, 0. So this is saying

f of x comma y is 0, 0 implies x comma y is 0, 0 that is the 0 in the vector space R2.

So, we have checked it's injective, so therefore, or one-one. So therefore, so this argument

is saying that it is f is one-one.

Let us check that it is surjective or onto. To check that it is onto we have to show that

if you are given any element let us say U comma V in R2, you can get x, y such that

f of x, y is U comma V. So, if so, for U comma V in R2 consider x to the U by 2 and y to

be V. Where did I get this?

This is not coming from magic, I want 2x to be equal to U and y to be V, so X must be

U by 2, and that clearly suggest that so therefore, f of x, y is 2x comma y is 2 times U by 2

comma V, which is U comma V. So, this is now saying it is onto. Great. So, we have seen

how to prove that this is an isomorphism.

So, we should also look at examples of things which are not iso, linear transformations

which are not isomorphisms. So, here is an example, which is not an isomorphism. The

second example that we saw. So, this is f of x, y is 2x comma 0. So, what goes wrong?

Well, several things go wrong. First of all, there is no pre-image for the vector U comma

V where V is non-zero.

For example, 01 has no pre-image. Why does not it have a pre-image because you can see

that the only vectors that are images are of the form something comma 0, so 0 comma

1 cannot be a pre-image, sorry, cannot have a pre-image. So that means there is no V in

R2 or no x, y in R2 such that f of x, y is equal to 0, 1 that is what we are saying.

So, this is not onto or surjective.

So similarly, if you take f of x, y to be 0, 0 then that means to 2x is 0, 2x comma

0 is 0 comma 0, which means x is 0, but y could be anything. So, for example, if you

take f of 0 comma 1, then the answer you get is 0. That means, we cannot say that f of

x, y is 0, 0 implies x, y is 0, 0 because you could have many other choices.

For example, you could have 0 comma 1. So, this is not one-one. So, both of the conditions

needed for a bijection fail. So, this is a, this is not an isomorphism by a margin. There

can be examples where you have one-one but not onto or onto but not one-one. Here is

an example which is not onto. So, f of t is t comma 3, t comma 23 by 89 t, this is one-one

but not onto, I will suggest you check why it is one-one but not onto.

And similarly, if you take the projection map R2 to R, then this is onto but not one-one.

Again I will suggest you check why it is onto but not one-one. It is clearly onto I think,

but 1, 1 again I think it is clearly not one-one. So, I hope, this gives you a feeling for what

we mean by an isomorphism.

So, let V be a vector space with basis V1, V2, Vn. So, now, let us introduce basis into

the picture. So, the what we are going to say here is that if you have a basis then

the linear transformation is determined by the values it takes on the basis. So, suppose

V1, V2, Vn is a basis and f is a linear transformation, then the ordered vectors f of V1, f of V2,

f of Vn uniquely determined f.

Why do I say this? So, the reason one says this is the following. If I want to get f

of V, I want to know what is f of V for some V in V. So, let v be indeed V, and I want

to know what is f of V. So, I am claiming that it is determined by f of V1, f of V2,

f of Vn. So, we know that if V1, V2, Vn is a basis, there is a unique linear combination corresponding to each which equals any given vector in that vector space. So V is summation

ci vi for a unique set of scalarsC1, C2 Cn.

So, then what is f of V? f of V is a f of summation civi, but now, I know that f is

a linear transformation. So, in particular I mean that means it satisfies linearity.

So, by linearity, I can pull out the summation and the scalars and write this as summation

ci times f of Vi. I do this sequentially. So, in my definition of linearity, I had f

of V1 plus C times V2 is f of V1 plus C times f of V2. So, I keep applying it to. First

I applied to Vn and then I applied to Vn minus 1 and so on. And I can so, I can pull out

any sum. It need not be only vectors it could be any finite sum. And similarly, I can pull

out scalars that is what the second condition of linearity told us.

So, since that is the case, this is determined, this f of V is determined by the values of

f of V1, V2, Vn. My values I mean the vectors, these are vectors, not numbers. So, if you take a vector V, and I want to evaluate f of V, all I need to know are what are the

scalars Ci and what are the vectors f of V1, f of V2, f of Vn, that determines f of V,

and that is what we mean by saying that a basis determines an ordered basis mind determines

the linear transformation.

I mean, we could flip this around, and say that suppose I have a, suppose I specify what

are I mean, some n many vectors f of V1, f of V2, f of Vn. So, suppose W1, Wn is a specified

set of vectors in W then there is a unique linear transformation if such that f of Vi

is equal to Wi. How do I define this linear transformation? That is exactly what this

equation here is telling.

So, what I do is I define f to be so define f of V is summation ciwi where V is equal

to summation civi. And then let us see what is f of Vi. So, f of Vi, so to get any f of

Vi, you have to first see how is Vi expressed in terms, how is that vector expressed in terms of the basis, but in terms of the basis Vi itself is a basis vector.

So, maybe let us say Vj, or let us say Vk. So, you put CK to be one, and all the other

cais to be zero, and that is the only way in which you can write, Vk as a linear combination

of these basis vectors, because it is a basis.

So, if you do that, then all the terms on the in this expression, submission, ciwi will

vanish except the kth one. And for the kth one, what you will get is CK is 1, so you

will get one times WK, which is exactly WK.

So, f of VK is WK, that is what we wanted. So now this linear, so this linear transformation

satisfies indeed that f of vi is equal to wi. And it is unique because every linear

transformation is determined by what values the basis vectors take. And since they take

I mean, these values are specified, no other linear transformation can force that f of

vi is equal to wi, that is exactly what we proved before. So with this, let us go ahead.

This is a slightly tricky point and I mean, I want a bit you to think a little about this

as we go ahead.

Let us do this on a particular example. Suppose I take the standard basis 1, 0, 0, 1 of R2,

what linear transfer. I should have pointed and said that second part that we did, where

we specified W1, W2, Wn and then we defined the linear transformation f that is called

extending the function f. So, we have f of vi is equal to wi and then we are extending

it to the entire vector space W sorry entire vector space V.

So here, in this example, we have, so W1, W2 are these two vectors 2, 0 and 0,1 and

let me write that so this is your W1 and this is W2. So we want to extend this function

f to a linear transformation from R2 to R2.

So, to do that, we have to write any x comma y first as a linear combination of 1,0 zero

and 0,1 X times 1,0 plus y times 01. And then I define f of x, y as x times f of 1, 0 which is 2,0 plus y times f of 0 comma 1, which is 0 comma 1. So, this is 2x, comma y. So

here, maybe I should put in a step there is 2x, comma 0 plus 0 comma y just 2x, comma

y. This was exactly the first example that we looked at, in this video of linear transformations

or linear mappings. So, this is the function that we get by or the linear transformation

we get by extending the function, which takes 1, 0 to 2, 0 and 0, 1 to 0, 1 that is the

way we say this.

So, let us change the basis and see what we get if we change the basis. So, the point

here is if you change the basis, you will get a different linear transformations, and

this is very important. So, in the previous example, instead of working with the standard

basis let us work with the basis 1, 0 and 1, 1. So let us say I take 1, 0 to 2, 0 and

1, 1 to 1, 1.

So, if that if you do that, then what do we get? So, every element x, y is uniquely represented in terms of this basis as x minus y times 1, 0 plus y times 1, 1. So, I think here,

there is a typo, this is f of 1, 1 should have been 0, 1 and not 1, 0.

We could, I mean, we can also do 1, 1 but it was intended to be 0, 1. So basically the

basis was changed, but the values were the same, so the W1, W2, Wn. So here we have W1

and W2, they are the same, but the basis V2, V2, Vn was changed. And the question is what

linear transformation do you get by extending? So let us apply the same thing.

So, if you have f of x, y, you write this as x minus y times f of 1,0 which is 2 comma

0, plus y times f of 1 comma 1. So, this is x minus y times 2 comma 0. So, note that W1

and W2 are the same. So, this part will, meaning these two things will remain the same. But

what changes?

What changes is are the coefficients this changes. So, y has remained the same, but

x has changed. So this is going to give us something else, this is going to give us two times x minus y comma 0 plus 0 comma y. So, this is 2x minus 2y comma y. that is the linear

transformation.

If instead we had chosen, I mean, just as an example, since I had that typo over there,

let me work that out as well. If instead, my basis was the one that I have here, 1,

0 and 1,1 and W1 was 2,0 and W2 was 1,1, then what would have x, y, have been? So f of x,

y would have been x minus y times 2 comma 0 plus y times 1 comma 1, if you work out

is 2x minus 2y plus y comma y, which is 2x minus y, comma y. So this is a second example.

So, let us just summarize what we have done this video. So, in this video, we have seen

what is the linear transformation. Essentially, it is just generalizing the definition of

a linear mapping, which was a bunch of linear combinations to arbitrary vector spaces. Of

course, in arbitrary vector spaces, we do not have the, I mean linear combinations,

we have to make sense of what that means.

So, instead of doing it that way, we specified by the other property, which is to say that

they are satisfying the two conditions of linearity or equivalently a single condition

that we call linearity in the previous video. So, either f of V1 plus V2 with f of V1 plus

f of V2 and f of C times V1 is C times f of V1 or equivalently, f of V1 plus C times V2

is f of V1 plus c times f of V2.

This is exactly the same, as saying that f of summation civi is equal to summation ci

times f of vi. This is linearity. So that is what was a linear combination, we saw,

of course, just by its very nature, that linear mappings are linear transformations. And we

saw a bunch of examples.

And we saw that they can be written in some kind of matrix form. And finally, we saw the

relevance of a basis to all this. Namely, that you can use the basis to express a linear

transformation in terms of the values that are taken by the basis vectors, and that actually determines the linear transformation entirely.

And finally, we saw that if you, you have to be careful about what basis you choose.

And so, if you choose different basis the linear transformations you are going to get

by extending the basis may be different, so that you have to be careful about. So, thank

you.